

Chemical Engineering (CEN1) Standard Operating Procedures (SOP) and DAC Notes, AY2024

I. Chemical Engineering Department Academic Counselors (DAC) Standard Operating Procedures (SOP)

This document outlines the responsibilities and expectations for DACs within the Chemical Engineering (CEN1) major. The DAC Notes portion describes recurring questions cadets ask of their DACs, as well as situations DACs need to be alert for within the Academy Management System (AMS). Although many of these issues show up during a Degree Audit or Grad Check, they are explained in more detail here. The Chemical Engineering Studies (CES1) major is discussed in the “Additional Notes” section of this document.

A. CEN1 DAC Responsibilities

DACs will be familiar with Figure 12 in the Notes section so that at any time that an interested cadet asks, they can discuss the general requirements for the major. DACs will also have a durable hard copy of the 8TAP available in their office for discussion. Additional information about the major requirements and courses can be found in the Chemical Engineering tri-fold brochure and/or the USMA Redbook.

To become a DAC, faculty need to request access to the “ADVISOR” app within the AMS system. This is accomplished through the AMS Applications and Modules function on the AMS homepage (Figure 1).

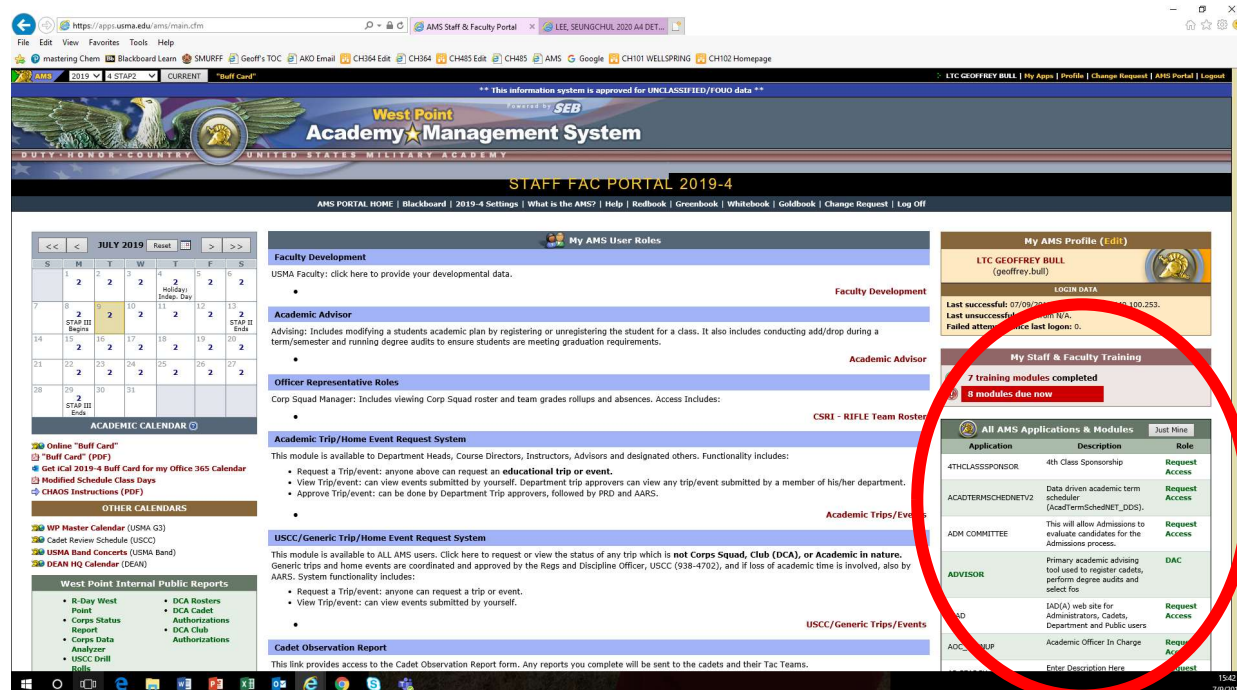


Figure 1. AMS Homepage with highlighted Applications section.

Once requested, contact the current Head DAC for the department who can grant new access. Access is usually granted within 24 hours. When access is granted, an Advisor can now be assigned cadets to advise within the major.

During the spring semester of their 4th Class (Plebe) year, prior to Spring Break, cadets will declare their major. As of AY2023 and beyond, cadets self-enroll in the major of their choice. The program leadership will be given a roster of cadets who have selected CEN1; any management of the final roster will be conducted by the program leadership. However, any cadet who is interested in the major and has questions, can inquire with any member of the program.

Once the roster of new majors is finalized, the program leadership will allocate new advisees to the program's DACs. First time rotating faculty will be allocated a smaller number of advisees than senior faculty. If a faculty member particularly wants to advise a specific cadet, this can be requested at any time.

NOTE: Any time an action is taken, or discussion is held (in person or electronically) between a DAC and cadet, an annotation of the action/discussion should be made in the "Advisor Remarks" in AMS.

The first requirement for DACs is to meet (in-person is preferred, but documented email conversations can be sufficient) with their new majors prior to the Friday of Week 13 of the semester (middle of April) and review the cadet's 8TAP. A synopsis of this meeting **MUST** be documented in the "Advisor Remarks" section of the 8TAP page (see Figure 2).

The first priority for each cadet is to identify any "Graduation Check Deficiencies." These will show up automatically when the 8TAP button at the top of the screen is clicked. An example is shown in Figure 2. The notes section describes many of the common fixes to deficiencies. The major exception is a cadet not having chosen their engineering electives. Resolution of this deficiency is described later.

Academic Plan

(Major-related courses are in **blue**)
Click on Major-related language courses to change course tag to CORE.
(n) denotes term course count for total academic classes having 3 or more credit hours

2019-1	2019-2	2020-0	2020-1	2020-2	2021-1	2021-2	2022-1	2022-2
(7)	(8)	(1)	(7)	(6)	(8)	(5)	(5)	(3)
EN101	EN102	ML100	CH102	CH362	CH363	CH364	CH365	CH400
HI105	HI106R		LR203	EV203	CH383	CH367	CH459	CH402
IT105	MA255		MS200	LR204	EE301	MC300	CH485	MX400
MA153	MD102		PE215	MA206	HI302	MC312	LW403	
MD101	MS100		PY201	MA364	MC311	SS307	PE450	
PE117	PE116		SS201	PE322	MS300			
PH255	PH256		SS202		PE360			
	PL100				PL300			
{5}	{5}	{0}	{5}	{5}	{6}	{5}	{4}	{2}
7	8	1	7	6	8	5	5	3

Graduation Check Deficiencies

MAJET UNDERLOADED:(18)
 COURSE COUNT FOR 2022 1 = 4
 COURSE COUNT FOR 2022 2 = 2
COURSE COUNT FAILED:(15)
 37 COURSES FOUND
FOS/MAJ TOTAL COUNT NOT SATISFIED:(24)
 HAS 18 OUT OF 19 Courses Required for Graduation
FOS/MAJ1 DIRECTED TRACK NOT SATISFIED:(20) [CEN1 details](#)
 Track Number: 2 Electives 0 out of 3 [register](#) | [waiver](#) (Note: request major course substitution waiver)
PRE REQUISITE FAILED FOR COURSE:(1) [course details](#)
 MA364

[Request Waiver](#) (Note: request any/all waivers)

Graduation Checks Satisfied

Advisor Remarks ([Click here to Edit](#))

No Remarks Found

Figure 2. Example 8TAP highlighting Grad Check deficiencies and Advisor Remarks sections.

The second priority is to run a “Degree Audit.” This button is found near the top of the 8TAP screen (Figure 3).

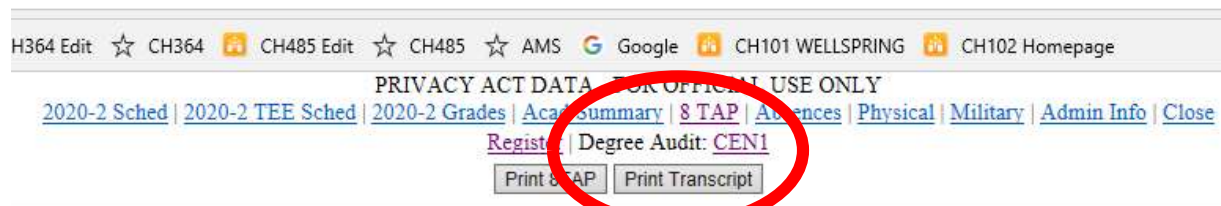


Figure 3. Example 8TAP screen highlighting the Degree Audit function.

The Degree Audit shows ALL courses that a cadet is required to take within the given major, in this case, Chemical Engineering (CEN1). These are broken down into various categories (major courses, cyber requirements, science depth, etc.) If the courses are in a cadet’s 8TAP, there will be a green **Y** in that category. If they have not been scheduled into the 8TAP, a red **N** will be present (Figure 4). This is your overall indication that a cadet’s 8TAP does or does not meet the requirements for the major.

[Degree Audit Course Edit](#) | [Close](#)

**DEGREE AUDIT
FOR
CDT DUFFY, BRIGIT A - USMA ID: C45958087 - CLASS: 2020**
Date: 07/10/2019 - Time: 10:41:25AM

[Refresh Page](#)

View Options: Default Mode - (All tracks and completed courses within each track) ▾ *Default Mode*

[Export to Excel](#)

MAJOR	TRACK	MAJ CRSE	MAJ CRSE TITLE	SATISFIED?	CMPLTD CRSE	SBST CRSE	ACAD YR	TERM
CEN1	Required Courses (12 of 12)			Y				
		1 CH362	MASS & ENERGY BALANCES		CH362		2018	2
		2 CH363	SEPARATION PROCESSES		CH363		2019	1
		3 CH364	CHEMICAL REACTION ENGINEERING		CH364		2019	2
		4 CH365	CHEMICAL ENG THERMODYNAMICS		CH365		2020	1
		5 CH367	INTRO / AUTOMATIC PROC CONTROL		CH367		2019	2
		6 CH383	ORGANIC CHEMISTRY I		CH383		2019	1
		7 CH400	CHEM ENG PROFESSIONAL PRACTICE		CH400		2020	2
		8 CH459	CHEM ENGR LABORATORY		CH459		2020	1
		9 CH485	HEAT AND MASS TRANSFER		CH485		2020	1
		10 MC300	FUND OF ENGR MECH AND DESIGN		MC300		2018	1
		11 MC311	THERMAL-FLUID SYSTEMS I		MC311		2019	1
		12 MC312	THERMAL-FLUID SYSTEMS II		MC312		2019	2
	Electives (3 of 17)			N				
		1 EE360	DIGITAL LOGIC W/ EMBEDDED SYS		EE360	ME201	2018	1
		2 MC380	ENGINEERING MATERIALS		MC380		2020	1
	Complementary Support Courses (3 of 5)			Y				
		1 EE301	FUNDAMENTALS OF ELEC ENGIN		EE301		2018	2
		2 MA365	ADV MATH FOR ENGRS/SCIENTISTS		MA365		2019	1
		3 PH256	ADVANCED PHYSICS II		PH256		2018	1
	Integrative Experience for the Major (1 of 1)			Y				
		1 CH402	CHEM ENG PROCESS DESIGN		CH402		2020	2
	Science Depth (0 of 0)			Y				
		1 CH102	GENERAL CHEMISTRY II		CH102		2018	1
	STEM Depth (0 of 0)			Y				
		1 MA255	ADV MULTIVARIABLE CALCULUS		MA255		2017	2
	IT/CYBER Requirement (0 of 0)			Y				
		1 CH367	INTRO / AUTOMATIC PROC CONTROL		CH367		2019	2
		2 EE301	FUNDAMENTALS OF ELEC ENGIN		EE301		2018	2
	Writing-in-the-Major (0 of 0)			Y				
		1 MC312	THERMAL-FLUID SYSTEMS II		MC312		2019	2

Figure 4. Example Degree Audit screen showing an incomplete 8TAP with a red “N.”

If courses are missing, in the wrong semester, or no longer required in the new major's 8TAP, the DAC will need to add them, move some around, or delete them. All these actions are accomplished through the REGISTER function in AMS, located as shown in Figures 5 and 6. A new major's 8TAP should be constructed to look as much like Figure 12 as possible.

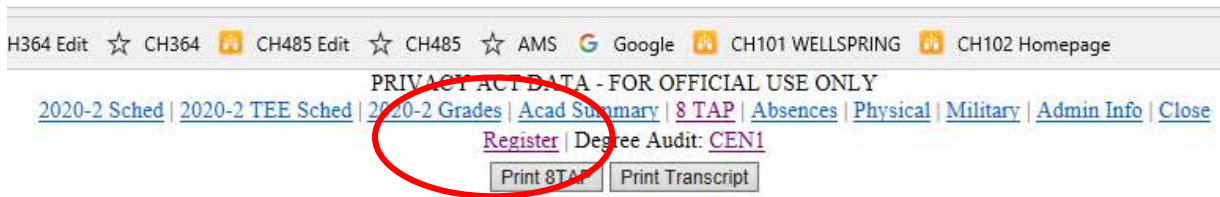


Figure 5. Location of REGISTER function in AMS.

The screenshot shows the 'Register' screen for a student named POLHEMUS, THOMAS LEONARD. The screen displays various fields for student information, including Name, Email, Cadet ID, Status, Class, Co, A1, 1st Major, 2nd Major, Eng Seq, TQPA, and CQPA. Below this are three main sections: 'Drop', 'Move', and 'Add'. Each section has a table with columns for Course, Acad Yr, Term, and a button. The 'Drop' section has a 'Drop' button. The 'Move' section has a 'Move' button. The 'Add' section has an 'Add' button. The 'Wait List' section has a 'Remove' button. The 'Drop', 'Move', and 'Add' buttons are circled in red. Below these sections is an '8TAP' table with columns for years and courses.

2019 1	2019 2	2020 0	2020 1	2020 2	2021 1	2021 2	2022 1	2022 2
CH101	EN102	ML100	CH102	CH362	CH363	CH364	CE350	CE450
EN101	IT105		HI108R	EV203	CH383	CH367	CH365	CH400
LS203	LS204		MA205	HI105	EE301	MC300	CH459	CH402
MA103X	MA104		MD201	MA364	MA206	MC312	CH485	LW403
MD101	MD102		PE215	MD202	MC311	MS300	HI302	MX400
PE117	MS100		PY201	MS200	PE360	PE450		
PL100	PE116		SS202	PE321	SS307	PL300		
	PH205			PH206				
				SS201				

Figure 6. A view of the REGISTER screen in AMS, showing the “Drop,” “Move,” and “Add” functions available to DACs.

Sometimes courses that should be automatically populated when cadets sign up for the major have been CLOSED by the hosting department. This happens for various reasons, but if they are part of the major, the DAC will need to place the cadet on the “Waitlist” in AMS, as shown in Figure 7.

PRIVACY ACT DATA - FOR OFFICIAL USE ONLY

[Cadet Detail](#) | [2021-2 Schedule](#) | Department: (Default - All) | Next Terms XRef | [Close](#)

Register

Name: POLHEMUS, THOMAS LEONARD **Cadet ID:** C52904335 **Status:** C **Class:** 2022 **Co:** A1
Email: thomas.polhemus@westpoint.edu **1st Major:** CEN1 **2nd Major:** **Eng Seq:**
AYT: 2020-2 **TQPA:** 3.202 **CQPA:** 3.194

This course has been closed by department.

Would you like to be added to a wait list for this course?

Remark:

Drop

Course	
2021 2 CH364	Drop

Move

Course	Acad Yr	Term	
2021 2 CH364	2021	2	Move

Add

Acad Yr	Term	Course	
2021	2		Add

Figure 7. A view of the “Waitlist” request screen in AMS.

In the remark, put something to the effect of “Chem Eng major, required course” or “Chem Eng major, requested elective.” You are limited to about 50 characters in the remark. Upon clicking “Yes,” the cadet will be on the waitlist, as shown in Figure 8. Waitlists are governed by the hosting department. The DAC who requests the course should be contacted within a few days by the host department. If not contacted within a week, you can find the Course Director of the course in the Redbook and query them on issues. It is possible that the department may not open the course for a particular semester for quite some time, but the discussion with the CD will clarify that. A cadet may have to be notified that they will not be able to take a particular course (if an elective).

PRIVACY ACT DATA - FOR OFFICIAL USE ONLY

[Cadet Detail](#) | [2021-2 Schedule](#) | Department: (Default - All) | Next Terms XRef | [Close](#)

Register

Name: POLHEMUS, THOMAS LEONARD **Cadet ID:** C52904335 **Status:** C **Class:** 2022 **Co:** A1
Email: thomas.polhemus@westpoint.edu **1st Major:** CEN1 **2nd Major:** **Eng Seq:**
AYT: 2020-2 **TQPA:** 3.202 **CQPA:** 3.194

CDT POLHEMUS, THOMAS L has been added to the wait list for EV301 (2022-2)

Drop

Course	
2021 2 CH364	Drop

Move

Course	Acad Yr	Term	
2021 2 CH364	2021	2	Move

Add

Acad Yr	Term	Course	
2021	2		Add

Wait List

Course	
2022 2 EV301	Remove

Figure 8. Confirmation that a cadet has been put on a Waitlist for a particular course.

The most common issue is that cadets have not picked and enrolled in their required 3 engineering courses outside the major. This discussion needs to be part of the initial counseling a cadet receives once they have declared the major. While cadets do not need to pick their engineering electives immediately, the earlier they choose them, the more flexible their 8TAP can potentially be. Electives can also be changed during normal “Open Enrollment” periods.

AT A MINIMUM, the Degree Audit function will be run once per semester during the week prior to open enrollment ending (usually week 13 – mid-April in the Spring, mid-November in the Fall). This update to the degree audit and any reported discrepancies will be recorded in the Advisor Notes section. An example is shown in Figure 9.

Advisor Remarks ([Click here to Edit](#))

REMARKS	
Degree Audit failed; needs another engineering elective. Also overloaded in 20-1; req out-of-cycle change to move MX400 to 20-2 to meet co-req. Emailed cadet about elective.	
Submitted By: LTC BULL (MADN-CHEM)	Date: 07/10/2019
Cadet desires to take more Arabic classes. Moved PE360 to pair with LTS in 19-2 and moved SS202 into 19-1 to accommodate this request. She may have to take an alternate lab to deconflict.	
Submitted By: DR LACHANCE (MADN-ORD)	Date: 05/02/2018

Figure 9. Example Advisor Notes indicating 8TAP deficiencies and actions taken by the DAC to resolve.

B. Waiver requests.

Waiver requests can be submitted for several reasons, including “Pre-Requisite,” “Co-Requisite,” and “FOS CRSE Substitution,” among others. These three are the most common within the Chemical Engineering major and of these, the FOS CRSE Substitution is by far the most common. Cadets often would like to take a course that has Engineering Topics (ET) credit that is not one of the “Templated” or “Pre-Approved” engineering electives that the AMS system has for the CEN1 major. This can happen because of new courses being added or just low-enrollment courses that our majors don’t usually take. The Chemical Engineering Program Director is the approval authority for any waiver request for Chemical Engineering majors.

The procedures for submitting a Course Substitution Waiver are as follows:

- 1: Identify which course your cadet would like to take.
- 2: Verify in the Redbook that the course meets the ET requirements (3.0) for the major.
- 3: Confirm with the Program Director that this course would be an acceptable engineering elective.
- 4: Enroll the cadet in the requested course, in the appropriate term, so that it appears on the 8TAP.
- 5: In the AMS 8TAP screen for the cadet, click the “waiver” link next to the “(Note: request major course substitution waiver)” description (Figure 10).



Figure 10. AMS 8TAP screen highlighting the waiver function.

6: In the next screen (Figure 11), you will see the newly enrolled course on the right-hand side of the screen. Select the requested course in this box.

7: The smaller box on the left lists all the APPROVED engineering courses. Select ANY of the courses in this box that the cadet is NOT enrolled in. This is the “Replacement” course.

8: In the “Remarks” section, be explicit in the terminology. In the example of Figure 11, this would be: “Request CH489 be substituted for EE377 to meet the engineering elective requirement. CH489 has 3.0 ET. Program Director approves.” Note: there is a 255-character limit to the Remarks section, so be concise.

9: Click the “Issue” button.

Cadet Details

FOS/MAJ Waiver Request
Deficiency Electives 2 out of 3
Deficiency Reason: 20
For: CDT DUFFY, BRIGIT Class 2020
Company H3

Waiver Type: FOS CRSE SUBSTITUTION (Code = 5)

FOS/MAJ Track Course List Replacement Course Number

EE377	2020 1 CH485
EM411	2020 1 CH489
EM420	2020 1 CH489
EM481	2020 1 LA475
MC306	2020 1 MC380
	2020 1 MC380
	2020 1 MX400
	2020 2 CH400
	2020 2 CH402
	2020 2 CH402

Remarks

Request CH489 be substituted for EE377 to meet the engineering elective requirement. CH489 has 3.0 ET. Program Director Approves.

Reset Issue

Figure 11. AMS Waiver Request example for course substitution.

10. The waiver will now be submitted in AMS. Inform the Program Director via email or in person that the waiver request has been submitted. The Program Director will approve the waiver via a note under Advisor Remarks for the cadet. This will log the date and time under the Program Director’s name.

11. Once the Program Director has approved the waiver in AMS, the Head DAC will issue an approval in AMS. The new course will become visible in the Degree Audit screen as one of the electives in the major.

Pending and Approved waivers are visible in AMS. However, still make note of the waiver request in the Advisor Notes.

II. Pre-Major-Selection DAC Notes

During their first semester at USMA, cadets are open to exploring different majors. They have a relatively short amount of time to make this decision and so very often ask questions about various majors. Especially within first- and second-semester General Chemistry classes, this provides an excellent opportunity to explain the benefits of the chemical engineering major. Aside from personal enthusiasm for the subject, there are a few ways to approach the major that can pique or enhance cadets' interest in the major.

A. Frequently Asked Pre-Major Questions

There are a few questions that cadets tend to ask routinely about majors in general, and specifically chemical engineering. Outlined here are some of them and some suggested ways of approaching answers to them.

1. Why choose chemical engineering?

I usually start with a summary of the Chemical Engineering major description found in the Redbook or in the program tri-fold brochure. This gives a good overview of things that Chemical Engineers do/can do.

2. What do you do with a chemical engineering major in the Army?

This one is a little tougher 'sell.' Some ways to approach; while there is no "Chemical Engineering Branch" in the Army, the 'Engineering thought process' or 'Engineering Design process' or 'Engineering approach to problem solving' is something that you can and will use in any branch of the military. This is why the Academy still requires a 3-course engineering sequence – so that all graduates will at least be exposed to that problem-solving process. Any of the engineering majors delves very deeply into this process, which can only help you to solve problems in the Army. It's not the only way, but as engineers, we think that it is a good, systematic way to approach problems.

Don't underestimate the possible experiences where your specific degree may be useful and needed. Some recent faculty anecdotes include:

Upon deployment, one of the first problems encountered with a local population was an inoperative water treatment plant. A cursory look around the plant showed an inoperative pump. Experience with the unit operations laboratory gives some insight to this sort of process.

A faculty member was specifically recruited to be part of an engineering team that was dealing with the production or destruction of certain chemicals of military significance.

3. Can I do a semester abroad?

See Section III.G

4. Can I do the pre-med track?

See Section III.H

III. CEN1 DAC Notes

Figure 12 is the Chemical Engineering major (CEN1) template 8TAP that should have been dropped in AMS when cadets declare their major. Unfortunately, sometimes the template does not work well and quite a few of our cadets will need some immediate attention in setting up a correct 8TAP.

4th Class Year		3rd Class Year		2nd Class Year		1st Class Year	
Fall Term	Spring Term	Fall Term	Spring Term	Fall Term	Spring Term	Fall Term	Spring Term
CH101	CH102	DFL1	CH362	CH363	CH364	CH365	CH400
EN101	EN102	MA205	EV203	CH383*	CH367	CH459	CH402
HI105	HI108	PH201	DFL 2	EE301	MC300*	CH485	Eng. Elective*
IT105	MA104	PY201	MA364/5	ME301	ME362	Eng. Elective*	LW403
MA103	PL100	SS201	PH202	MA206	SS307	Eng. Elective*	HI302
			SS202	PL300			MX400
{5}	{5}	{5}	{6}	{6}	{5}	{5}	{5}
17.0	17.5	23.0	20.5	17.0	16.0	16.5	17.0
Can be taken either semester of the indicated year		CEN1 Exception to Policy -Flexible		CEN1 major courses – generally MUST be taken in the indicated semester *most flexible courses			

Figure 12. Standard CEN1 8TAP.

The highlighted blue rows show the total number of academic courses and credit hours (respectively) taken each semester. They obviously are not balanced and when validations and potential research credits are accounted for, this template will not serve our majors well in all cases. The notes below indicate the limits within which DACs should operate when advising class of 2024 and beyond cadets. Additionally, Figure 12 does not include physical educational program requirements; these are managed by DPE.

A. Overall:

Validations and research aside, CEN1 cadets should have 2 semesters with 6 courses and 6 semesters with 5 courses. Except in cases requiring program director approval (e.g., double majors), all CEN1 majors will have at least 52.0 credits of engineering (ET).

Make routine use of the Degree Audit feature in AMS. This feature generally catches things like pre/co-requisite failures. Sometimes there are 'issues' that really aren't, but the system hasn't been alerted to them. For example, it may give an error for MA205 prerequisite for CH362 but the cadet took MA255 (advanced version). If you get this sort of error, just let the program Head DAC (LTC Cowart for AY2024) know, or contact the Registrar's office directly. The POCs in Registrar's office are Mr. Bob Gosicki or Dr. Russ Lachance.

Make generous use of Advisor Notes feature in AMS. Any changes, requests, etc., need to be documented. **Per the SOP, at a minimum, each cadet should have one note per semester, indicating Degree Audit was run.**

B. Constraints:

CH102 and MA205 must be taken NLT fall term 3rd class year (program director approval for spring term)

MA206 can be taken first semester 2nd class year.

CH362, MA364, CH363, CH364, CH459, C365, CH485, CH400, CH402 must remain in their respective semesters as listed.

CH383, ME301, ME362 can be moved forward (earlier) but not later. ME301 and ME362 can be taken either semester of the 2nd class year.

DFL1, DFL2, PY201, SS201 (economics), SS202 (American Politics) remain in 3rd class year (unless taken earlier; can be either semester).

PL300 and SS307 remain in 2nd class year (can be either semester).

LW403 remains in 1st class year (either semester).

HI302 is (as of AY2024) a co-requisite for MX400. MX400 cannot be taken first semester of 1st class year if HI302 is in the second semester.

C. Branches

The "Jedi Math" track involves cadets usually validating MA104. This involves MA153, MA255, and then MA365 in lieu of MA364. For cadets that ask, the bulk (75-80%) of MA364 is covered in MA153/255. As a result, the remaining 20-25% is still covered in MA365, but additional topics, such as some linear algebra, are covered, to avoid 1) double teaching content and 2) cadets getting bored.

D. Flexibility

HI302 can be taken 1st class year (either semester); this will happen often.

MA206 can be taken 2nd class year (usually fall term; program director approval for spring term).

Engineering electives can be taken any semester. For the most part, engineering electives can be any course with 3.0 or more engineering credits (ET) but require program director approval if a degree audit does not recognize them.

SS201 (Economics) may be moved to fall term 2nd class year with program director approval (not common).

MC300 can be moved forward or as late as first semester 1st class year.

E. STAP

Voluntary STAP has become much more 'acceptable' in recent years and is a way that cadets can free up time in their schedules for other courses (e.g., SAP/SAEP, med school option). Generally, these are core courses, and cadets must make sure that they are meeting summer graduation requirements on the Comm's side (Buckner, CLDT, CTLT, military IAD, etc.). Commonly available STAP courses are: MA206, PL300, PY201, sometimes SS201, SS202, or SS307, and sometimes HI302 (not common).

F. Additional courses

Many chemical engineering cadets will validate one or more courses. Because validations now count as credit hours received for accreditation purposes (Redbook, Part 1 "The Academic Program"), the validation of more than 2 courses will require the selection of electives beyond the three required engineering electives. While no constraints are ultimately placed on these electives once cadets have met the requirements of the major, some courses of action are preferred to others. Additional guidance follows:

1) If GPA is appropriate (generally > ~3.3-ish, and if they meet the criteria for the honors major), try to guide toward one or more semesters of 3.0 credits of research (CH489/490/491/492).

2) Recent classes (and some advisory board members) have pushed for adding the second semester of organic chemistry, CH384. If the cadet is willing, go for it but it should really be added in during the semester immediately following the one in which they take CH383. This may result in scheduling issues.

3) Collectively, our (and most) cadets struggle with fundamental engineering math skills. In the past, the Linear Algebra (MA371) has been worthwhile. Most classes involving differential equations are probably worthy electives. Other complementary courses are MA391, MA396, CH481/2.

4) Other chemistry courses that have long-term benefits for chemical engineers: Introduction to Analytical Chemistry (CH371) and Polymer Chemistry (CH471) are both recommended. Any other basic chemistry (Physical Chem, Inorganic, etc.) could all be acceptable. These are all time-intensive lab courses.

5) Cadets could take additional language courses, or they could choose a minor. Encourage them to have a goal toward which these additional courses will help them.

G. Semester Abroad Program (SAP) / Service Academy Exchange Program (SAEP)

We continue to get routine requests for either SAP or academy exchange among our majors. The overarching limitation for our majors is that we only offer our chemical engineering mandatory courses each in a single semester. As a result, there are two major considerations to pass along to your cadets who are interested in SAP or SAEP:

1. USMA is the only service academy with chemical engineering as a major. This does not preclude a cadet from doing SAEP, but they will be required to take our chemical engineering requirements **in addition to** any other service academy requirements they incur. This must be done remotely while they are at another service academy.
2. For semester abroad, there are two options: attend SAP at an institution with an ABET accredited chemical engineering program and take courses that correspond with their 8TAP for the term in which they are abroad. Alternatively, the cadet may be able to take their USMA courses remotely while abroad.

As a program, we are currently offering only CH363 in a remote format during the fall semester of their 2nd Class (junior) year, subject to the consideration of the CH363 course director. That format basically requires cadets to learn much of the material on their own via the textbook. Once a week, they and the instructor will link up in Teams for specific face-to-face questions, etc. Very often, they are back at USMA in time for the TEE, which they will take, either before the winter break or during the spring semester Reorgy Week.

For other required courses during the fall of the 2nd class year, the parent program may or may not allow the remote option. As an example, cadets will have to take ME301 or ME362 via distance during the fall of their 2nd class year. CME has not approved this option during the past few academic years. Additionally, the organic chemistry course is also problematic due to the lack of organic lab as part of another institution's organic chemistry course. This has led the chemistry program to not approve another institution's organic chemistry offering abroad. Since CH383 is only offered during the fall term, the best option is for a cadet to take CH383 as a yearling. Otherwise, they will be taking it during the fall of their firstie year (not recommended).

The bottom line on SAP/SAEP is that the only term for which the chemical engineering program will entertain the possibility of sending a cadet abroad is the fall of their 2nd class year. No other options are feasible without putting the cadet at significant risk of graduating on time.

H. Medical School Option

Generally, we get 1 or 2 chemical engineering majors that ask about the possibility of also doing medical school. There are a couple of standard responses.

First, try to have this conversation with cadets that express interest at the time of their signing up for the major. Generally, approach it from the perspective of what their long-term goals are. The two semi-standard cadet responses are:

- (1) I want to go to medical school or,
- (2) I want to be an engineer, but I want to hedge my bets.

In case (1), the general guidance is that it would serve them best to be part of the program that would set them up best for medical school in the long run, the life science major. However, this does not guarantee them a medical school spot (only about half or less of life science

majors end up going to graduate school). Absent medical school, an engineering degree is far more valuable than life science in terms of job prospects. If they express such interest as bioengineering or some other engineering based long-term goal, the engineering degree is also more appropriate. If they acknowledge and understand these ideas, then case (2).

In case (2), it can be a challenge to get the schedule to work out. However, there are some general guidelines and discussion points below.

If they don't have at least 2 validations, it's going to be VERY difficult. If they haven't taken CH102 in the spring of plebe year, then it is likely impossible to get their schedule to work without being significantly overloaded; they really need to take Orgo 1 and 2 (CH383/384) in yearling year. The other courses they need to take (mandatory) are: CH375 Advanced Biology, CH387 Human Physiology; and CH473 Biochemistry. All of these need to be taken by Spring of their Cow year so that they have them on their transcripts during the medical school application process, which begins that semester. They also need them for taking the MCAT, generally June after their Cow year.

The cadets need to get linked in with the Pre-Medical Club organized by the Life Science major. Some other Med School notes, that should be discussed in the pre-med club meetings, but probably need to be highlighted are:

Medical schools apparently like to see research on the transcript. This does not have to be medical or bio-related, but just some research. Generally, recommend having a 400-level research course on the books prior to the application period (i.e., by spring of Cow year).

They need to have some sort of 'physician shadowing' experience on their resume. This does not have to be via an AIAD or anything formal, they just need to demonstrate that they have had some in-depth interactions with a physician, preferably more than one, to show they are serious about this process. This is of concern, not so much for the medical school application, but to get them past the internal West Point vetting process.

The Dean's Office POC for medical school matters is Dr. Al Beitler, x6328.

I. Additional Notes

When new cadets are assigned to the chemical engineering major, the majority of the 8TAP will be auto populated with the required courses. However, several courses will not populate and will need to be manually added. CH362 routinely is not added due to the course being closed for registration. This allows us to control who takes the course; several other majors' cadets want to take the course, but we need to ensure chemical engineers are in the course first. Additionally, CH383 must be manually added for the same reason. It is highly recommended that all chemical engineering cadets take CH383 in the fall of their yearling year to help alleviate scheduling issues in future terms. This will only work if the cadet has taken CH102 in the spring of their plebe year; this does not always happen.

Enrolling cadets in STAP is a great way to mitigate scheduling issues for our cadets. STAP sign-up forms are available on the AARS [SharePoint site](#), however voluntary STAP is usually given lowest priority. Ensure that your Corps Squad athletes make good use of STAP when they can.

Ensure that you keep an eye on your advisee's 8TAPs regularly (refer to the instructions in paragraph I.A.). We have had instances of AARS moving our cadets' courses to different terms without warning. These movements should generate email notifications to the advisor; ensure that you pay attention to schedule changes; these changes can have implications for ABET accreditation if we have cadets taking courses out of sequence and generating prerequisite violations. Routine maintenance of your advisee's 8TAPs is necessary.

We also have a chemical engineering studies (CES1) major that is available; you may find yourself advising a cadet who is actually in CES1 for several reasons. A cadet may fail several courses, necessitating a change to their requirements to alleviate schedule pressure; a cadet may want to pursue the medical school option and need the schedule space. The major difference between the chemical engineering studies degree and the normal chemical engineering degree is that the studies degree is not ABET accredited, so does not have the same requirements. Most of these DAC notes still apply, but the 8TAP is slightly different. Refer to the Redbook for more information, but the following courses are not required for the CES1 major: CH400, CH459, and CH485. Additionally, the Writing-in-the-Major (WiM) requirement shifts to CH365. Note that CH402 is still required for the CES1 major.